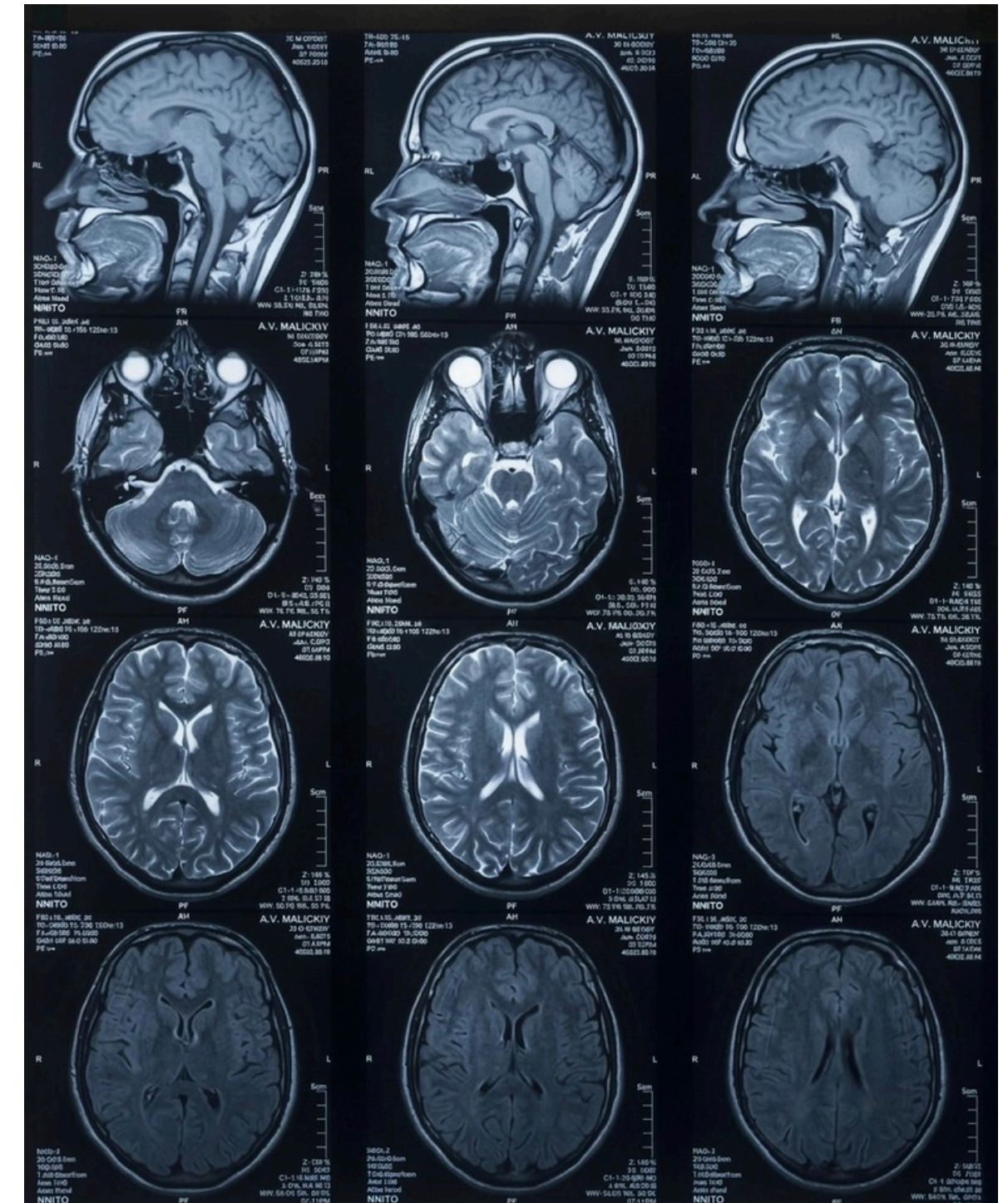




BRAIN-TISSUE SEGMENTATION

from unsupervised approach to deep learning

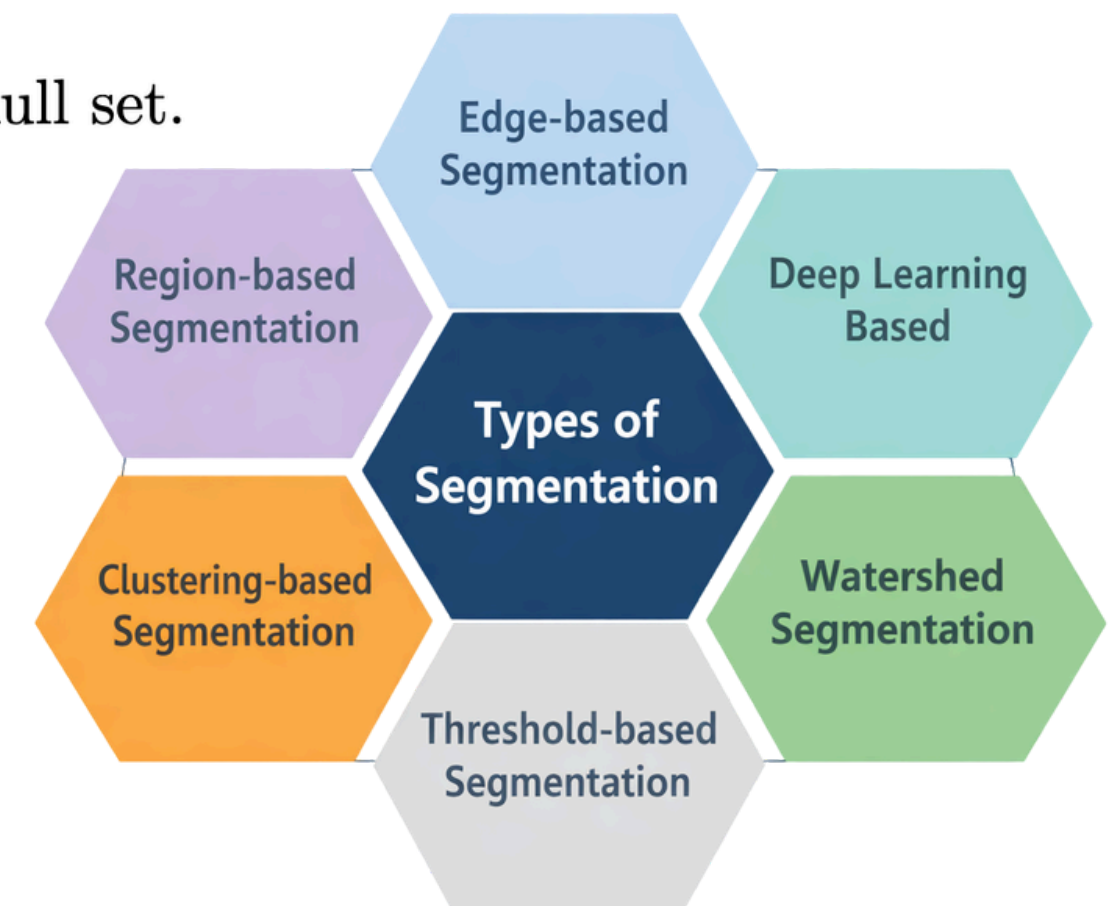


WHAT IS IMAGE SEGMENTATION?

Let R represent the entire spatial region occupied by an image. We may view image segmentation as a process that partitions R into n subregions, $R_1, R_2, \dots, R_n$, such that:

- (a) $\bigcup_{i=1}^n R_i = R$.
- (b) R_i is a connected set, for $i = 1, 2, \dots, n$.
- (c) $R_i \cap R_j = \emptyset$ for all i and j , $i \neq j$.
- (d) $Q(R_i) = \text{TRUE}$ for $i = 1, 2, \dots, n$.
- (e) $Q(R_i \cup R_j) = \text{FALSE}$ for any adjacent regions R_i and R_j .

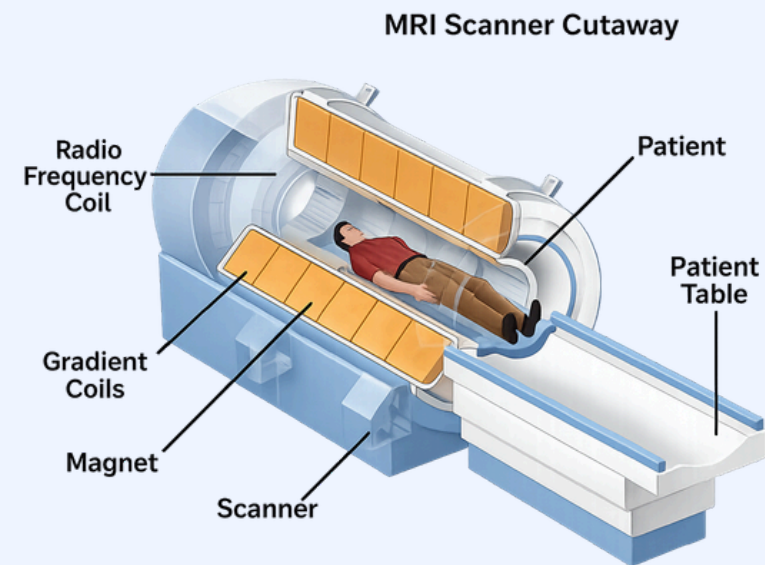
where $Q(R_k)$ is a logical predicate defined over the points in set R_k , and $\emptyset$ is the null set.



MAGNETIC RESONANCE IMAGING

Acquisition

A magnetic field combined with a radio-frequency pulse causes hydrogen protons to tip out of equilibrium.



Reconstruction

The emitted MR signals are collected in k-space and then mathematically converted to spatial images via Inverse Fourier Transform:

$$I(m, n) = \frac{1}{MN} \sum_{k_x} \sum_{k_y} S(k_x, k_y) e^{i2\pi \left(\frac{k_x m}{M} + \frac{k_y n}{N} \right)}$$

Image contrast

Clinically relevant contrast types are T1-weighted and T2-weighted imaging, governed by: $S \propto \rho \cdot \left(1 - e^{-TR/T_1}\right) \cdot e^{-TE/T_2}$

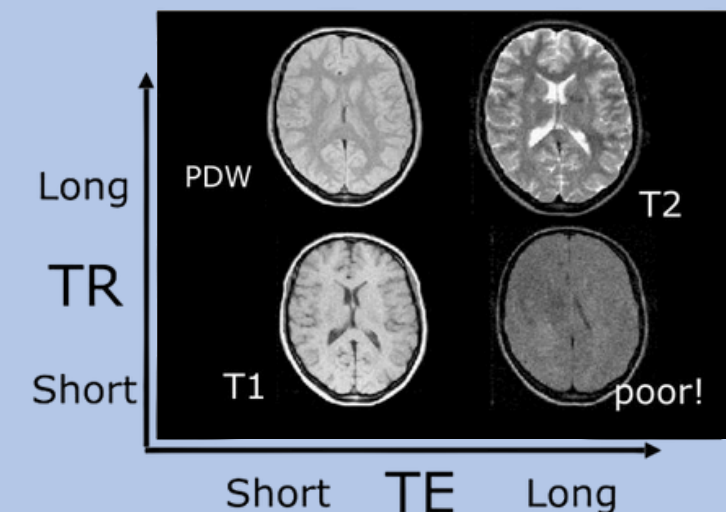
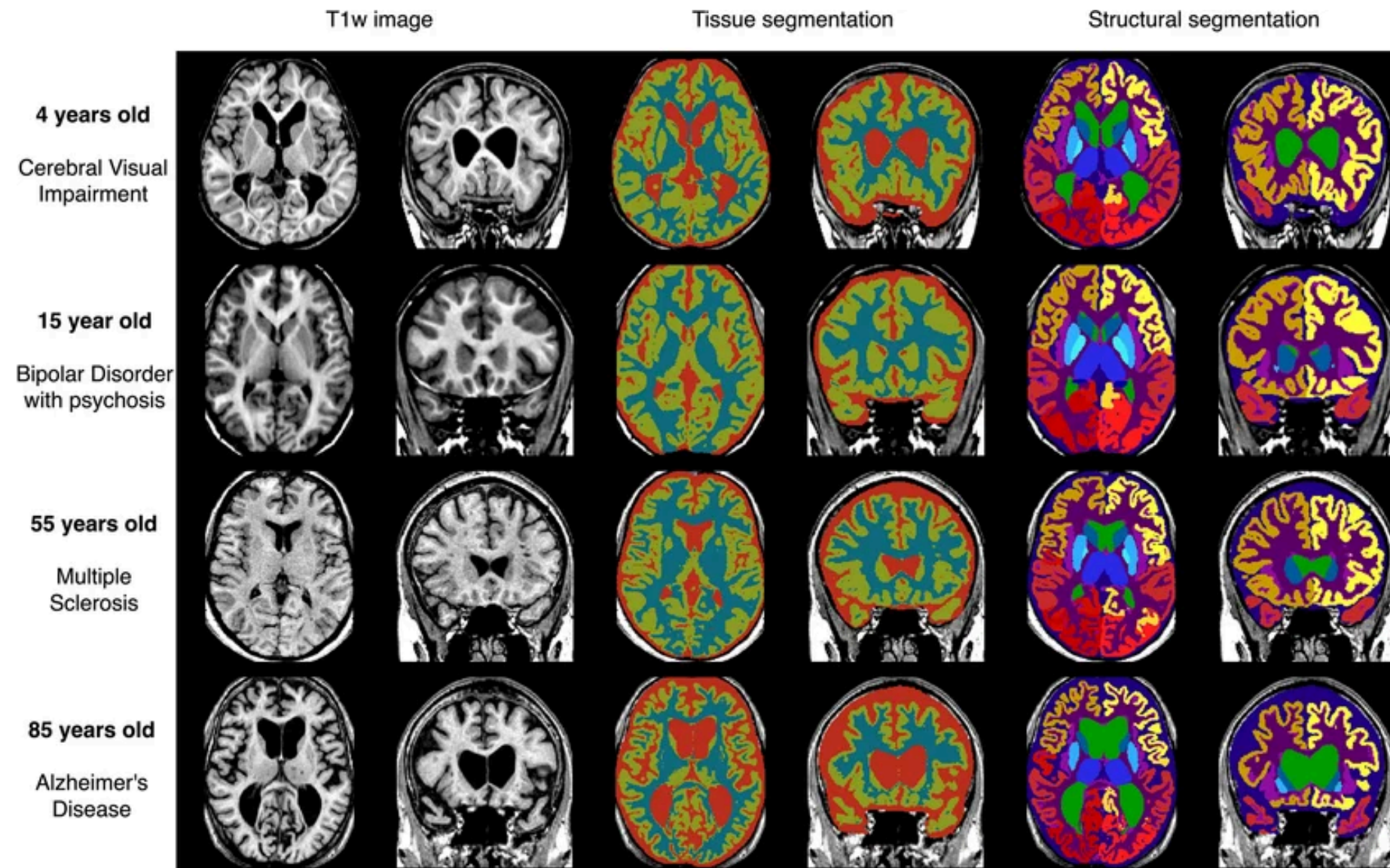


IMAGE SEGMENTATION IN BRAIN IMR

- Its goal is to partition the image into meaningful, non-overlapping regions by assigning a label to each voxel, transforming raw intensity data into a structured anatomical representation that can be quantified, compared, and interpreted clinically.

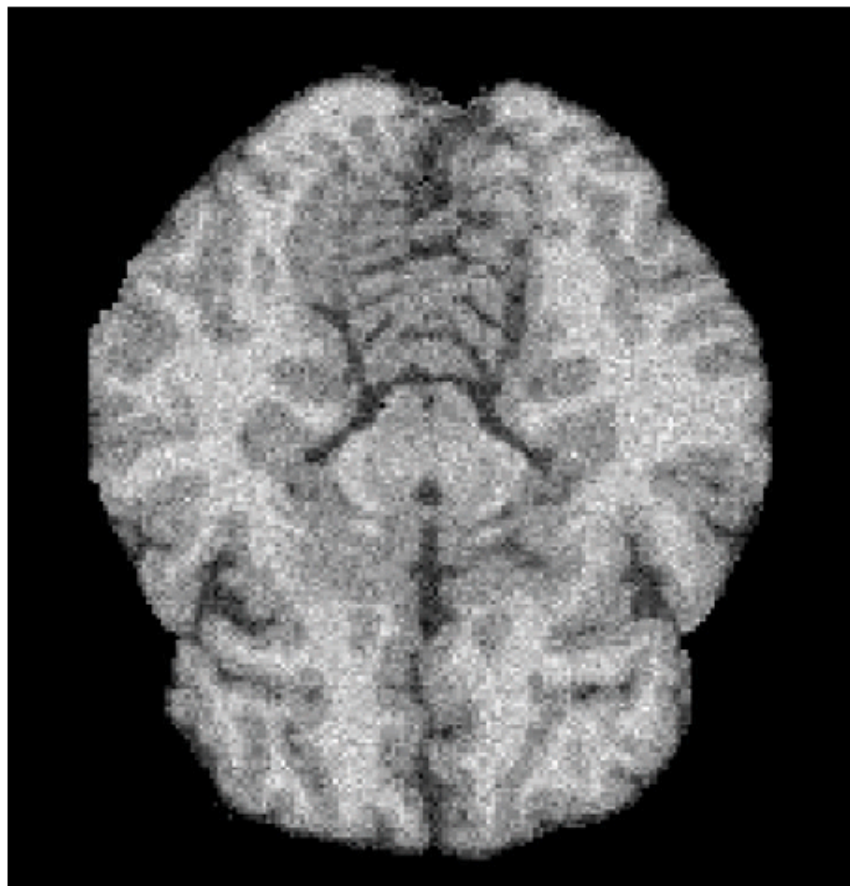


- Segmentation can be organized into:
 - **Tissue:** classifies the brain into its fundamental biological components, primarily White Matter (WM), Gray Matter (GM), and Cerebrospinal Fluid (CSF).
 - **Structural:** delineates specific anatomical sub-structures, such as the hippocampus, thalamus, amygdala.
 - **Pathological:** targets abnormal tissue variations within the brain, such as glioblastoma tumors, multiple sclerosis lesions, edema, or ischemic strokes

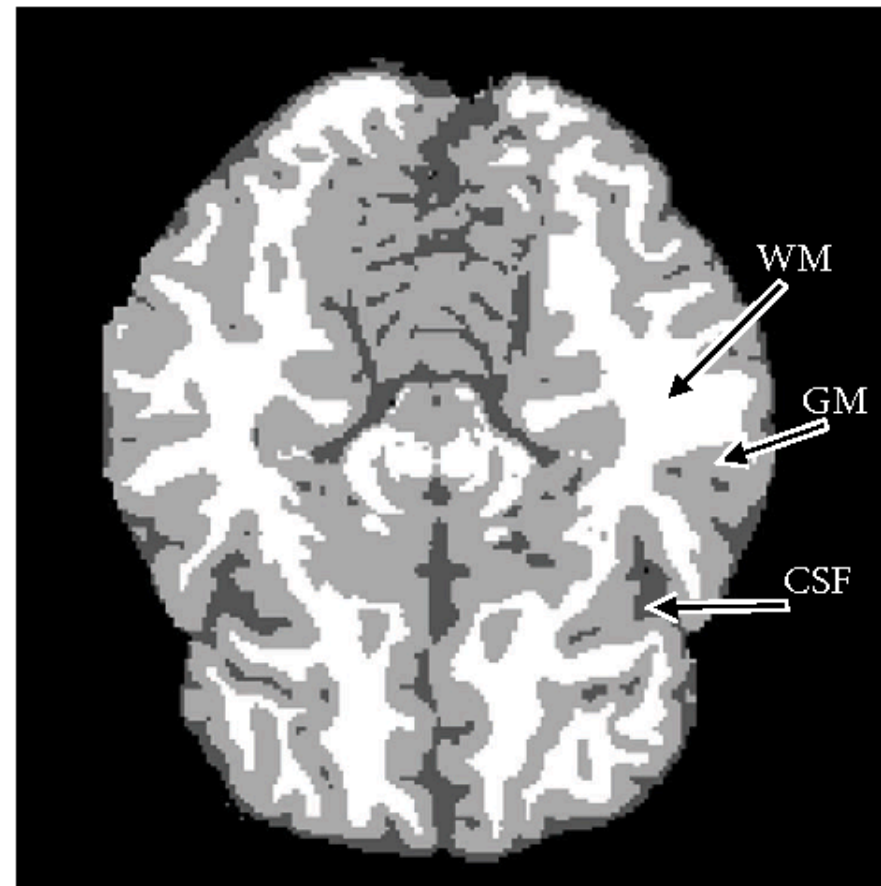
BRAIN TISSUE-SEGMENTATION

- Brain tissue segmentation is one of the most studied problems in neuroimaging. Its clinical importance stems from the tight relationship between tissue composition and brain health:
 - Gray matter volume → cognitive decline, Alzheimer's, Parkinson's
 - White matter integrity → multiple sclerosis
 - CSF distribution → hydrocephalus, atrophy

Original MR image



Segmented image



- What makes the brain tissue segmentation particularly challenging is the nature of the MRI signal itself.
 - GM and WM have overlapping intensity distributions in T1w images, boundaries are rarely sharp
 - Scanner noise, magnetic field inhomogeneities, and inter-subject anatomical variability mean no single intensity value reliably identifies a tissue class across different subjects or acquisition protocols

UNSUPERVISED APPROACH: K-MEANS

How it works

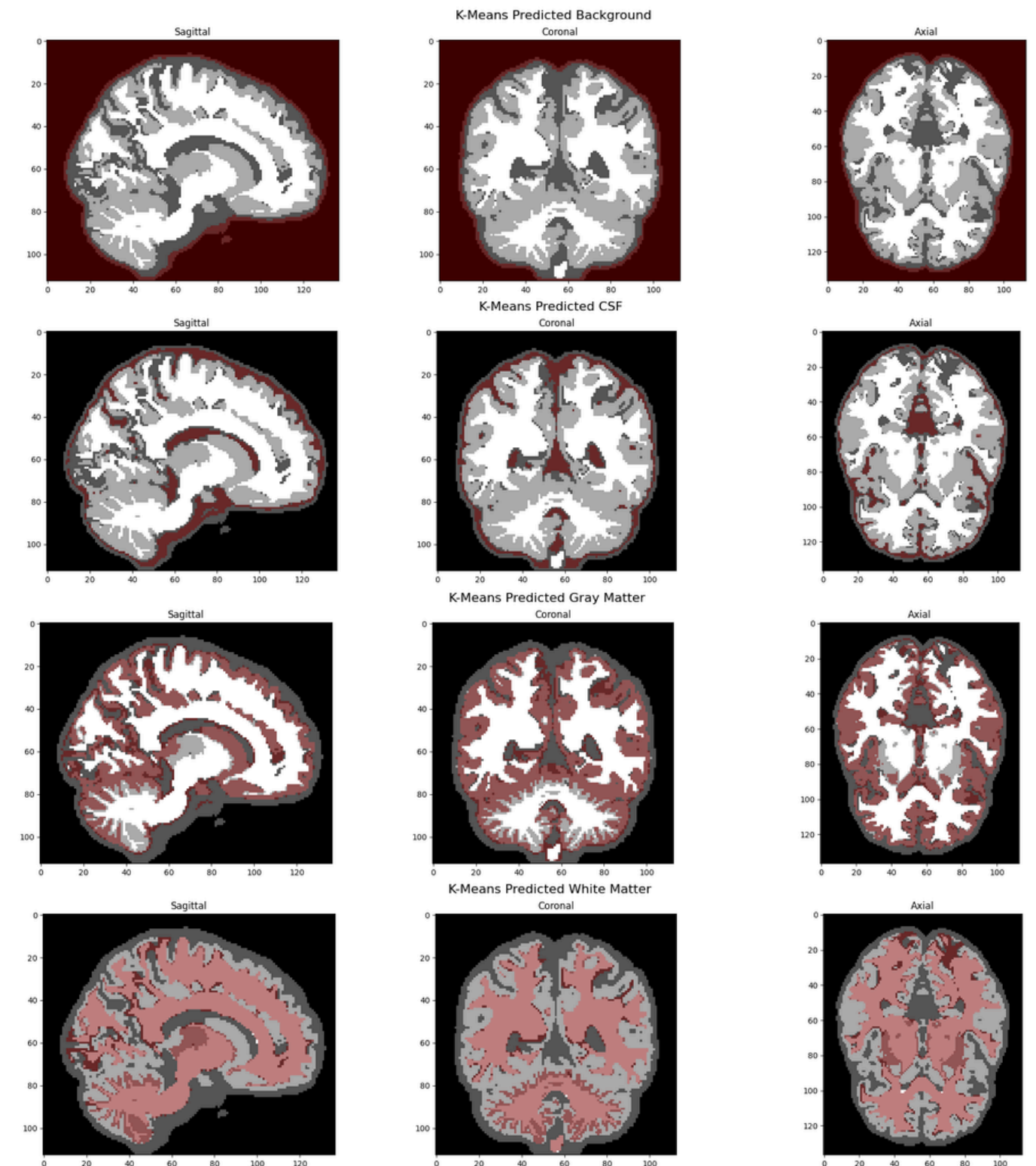
Each voxel is treated as a 1D data point whose only feature is its intensity value. The algorithm partitions all voxels into $k = 4$ clusters by iteratively assigning each voxel to the nearest centroid and updating centroids until convergence, minimizing the total within-cluster variance.

Label assignment

Since cluster indices carry no semantic meaning, an optimal mapping is applied post-clustering to match predicted clusters to tissue classes by maximizing the Dice score against ground truth.

Strength vs Limitation

No annotated data required: the model relies solely on raw intensity values, without any supervised information.



SUPERVISED APPROACH: U-NET

What it is

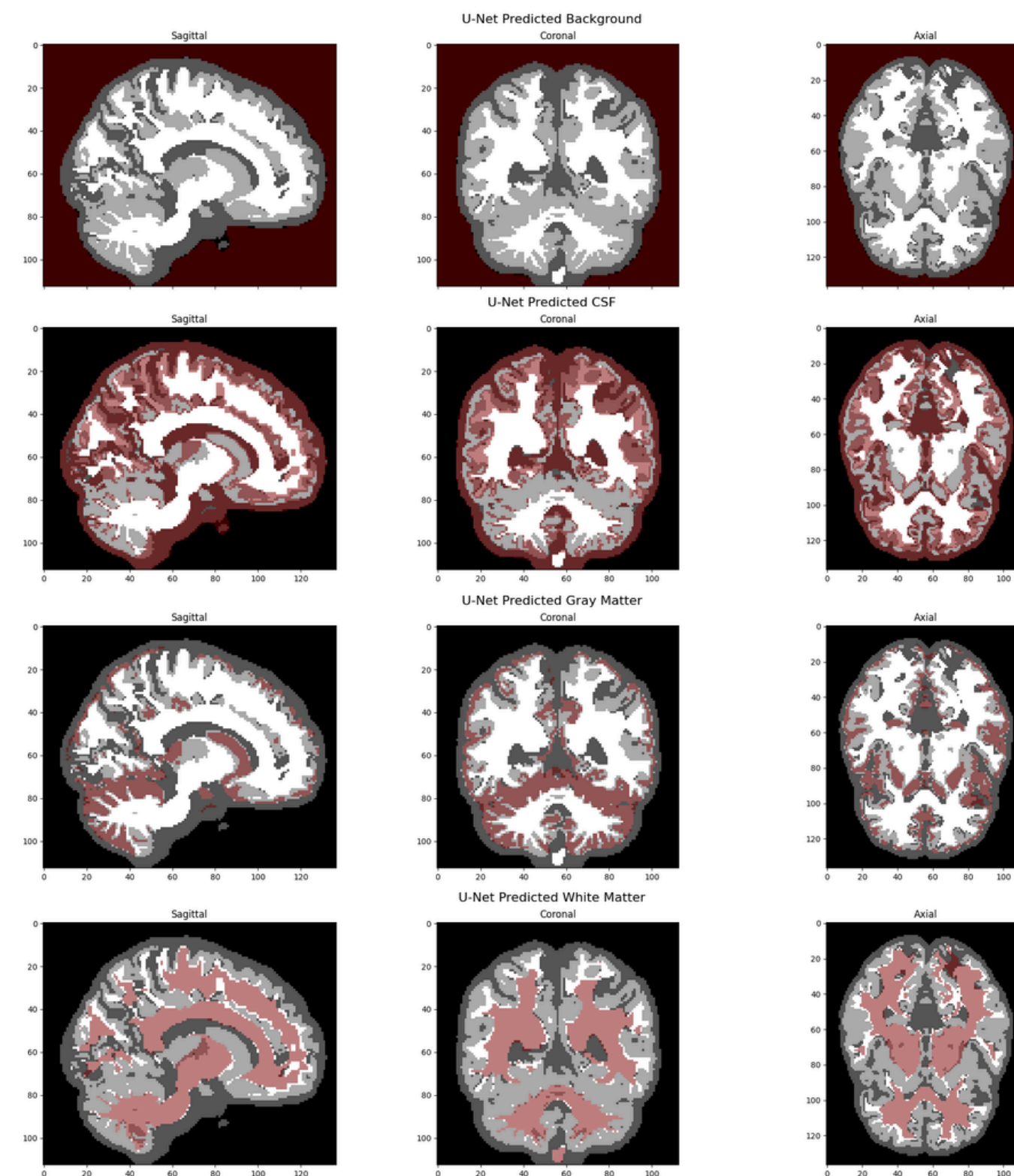
A fully convolutional encoder-decoder network extended to 3D volumetric data, with a single skip connection at the bottleneck. Unlike K-Means, it learns tissue patterns at multiple spatial scales, combining local intensity with broader anatomical context.

Implementation choices

To keep the model computationally tractable, a lightweight architecture was adopted and training performed on $64 \times 64 \times 64$ patches rather than full volumes. At inference, a sliding window strategy with Gaussian weighting reconstructs the full volume.

Training

The model takes a single-channel 3D patch as input and produces a 4-class voxel-wise segmentation map trained with cross-entropy loss over 30 epochs.



EVALUATION & CONCLUSIONS

CSF - cerebrospinal fluid

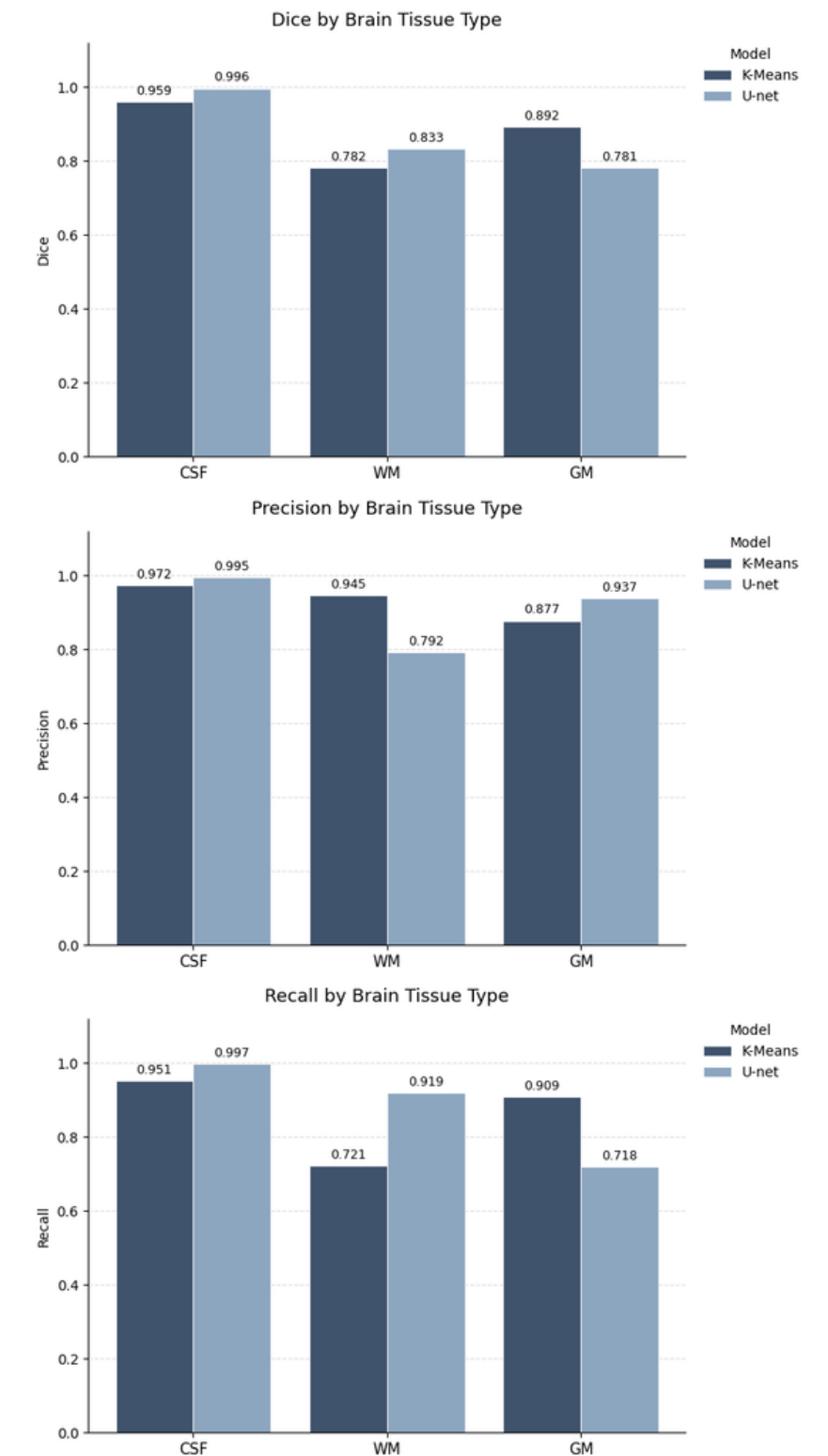
Both methods perform strongly; U-Net marginally better (Dice: 0.996 vs 0.959). This is expected: CSF occupies a distinct low-intensity region, making it the easiest class to separate.

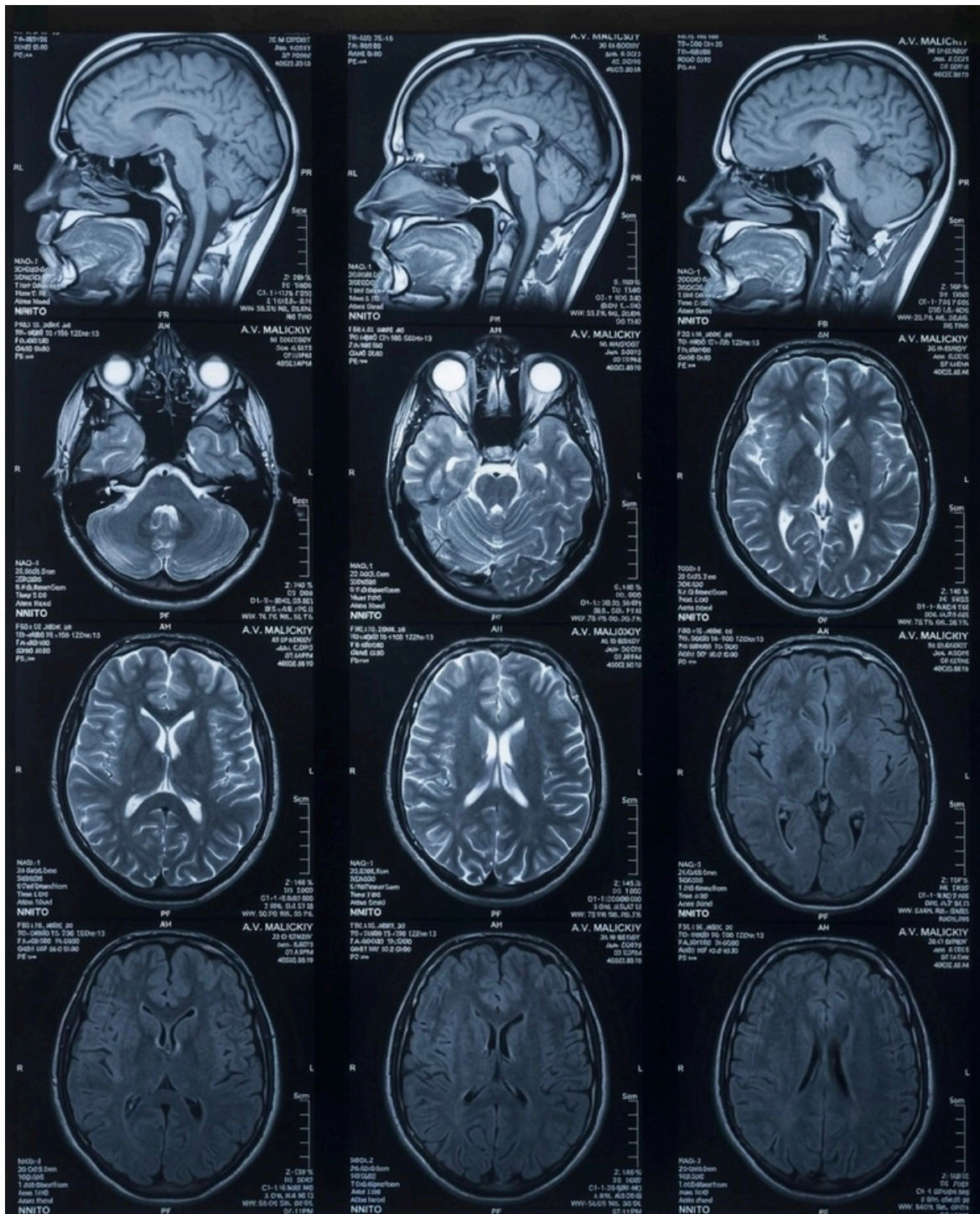
WM - white matter

U-Net leads in Dice (0.833 vs 0.782) and Recall (0.919 vs 0.721), recovering more true WM voxels. K-Means achieves higher Precision (0.945 vs 0.792), being more conservative in its predictions.

GM - gray matter

K-Means surprisingly outperforms U-Net in Dice (0.892 vs 0.781) and Recall (0.909 vs 0.718). The U-Net, trained on a limited dataset with a simplified architecture, struggles with gray matter's spatial complexity.





THANK YOU FOR
YOUR
ATTENTION